

Abstract

Deep Reinforcement Learning (DRL) controllers for HVAC systems are typically trained against neural-network surrogates because high-fidelity physical simulators are computationally prohibitive for the millions of environment steps required by policy-gradient algorithms. A natural assumption in building energy modeling is that a predictively superior surrogate yields a more effective training environment. We systematically test this assumption on the BOPTEST `bestest_air` testcase and report a critical negative result: the *fidelity–utility paradox*.

A physically structured grey-box surrogate (v3.5) utilizing a Neural Ordinary Differential Equation (Neural ODE) backbone with an identified zone thermal capacitance of $C_{zon} = 4.413 \times 10^5$ J/K achieves a 24-hour rollout RMSE of 0.644 °C, significantly outperforming a black-box control-oriented surrogate (v3, 1.557 °C rollout RMSE) in offline predictive accuracy. However, when v3.5 is used directly as the training environment, the resulting policy collapses to a maintenance score of $m_s = 1.046$ on the live BOPTEST Runtime Environment (RTE), exhibiting comfort violations above 77% and a live RMSE exceeding 4 °C.

Conversely, training on the predictively inferior v3 surrogate yields a highly functional policy, achieving $m_s = 0.073$ on the peak window and $m_s = 0.095$ on the typical winter window. We resolve this paradox by establishing a hybrid training architecture in which v3 provides smooth gradient fields for policy rollouts, while the calibrated v3.5 physical twin acts as a frozen per-step reward-shaping censor.

Subtracted disagreement penalties with $\lambda_{temp} = 0.10$ and $\lambda_{pwr} = 5 \times 10^{-5}$ enable the hybrid backend to sustain 1,786.8 steps/s on a single CPU thread, corresponding to an 85.0× acceleration relative to BOPTEST, while achieving $m_s = 0.087$ on the peak window and $m_s = 0.041$ on the typical winter window with comfort violations below 5%.

We further demonstrate that the optimal regularization strategy is architecture-specific. Thermostatic PPO requires $\lambda_{temp} = 0.10$, whereas Hierarchical Reinforcement Learning (HDRL) and 17-dimensional Multi-Objective Reinforcement Learning (MORL) achieve optimal performance with $\lambda_{temp} = 0.00$, indicating that power-only regularization is sufficient.

A pre-registered transferability study, identified by Git manifest anchor `1861e48`, validates the framework across three distinct hydronic systems. The proposed Stage A/B/C inverse calibration pipeline generalizes robustly, providing RMSE reductions of 60.2%, 87.4%, and 87.8%, respectively, and consistently re-identifying the zone thermal capacitance at 1.918 ± 0.032 times the baseline value despite an order-of-magnitude variation in zone volume.

Nevertheless, frozen-policy transfer remains strongly regime-dependent. Residential hydronic testcases fail the $1.25 \times$ PI comfort safety threshold, whereas the commercial stretch testcase satisfies the safety criterion but incurs a 35.3% energy penalty relative to the baseline controller. All numerical findings are linked to pre-registration audit anchors and fully comply with the Hou–Evins Level-3 reporting guidelines.